

Part b Verification of FastAPI - Improve the documentation (documentation of the function body)

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ uvicorn main:api --reload
INFO: Will watch for changes in these directories: ['/home/ubuntu/datascientest-data-engineering/fast-api']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [1912] using StatReload
INFO: Started server process [1917]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

Edit [main.py](#) file:

```
main.py M X
datascientest-data-engineering > fast-api > main.py > ...
1  from pydantic import BaseModel # type: ignore
2  from typing import Optional
3  from fastapi import FastAPI # type: ignore
4
5
6  api = FastAPI(
7      title = "My API",
8      description = "My own API powered by FastAPI.",
9      version = "1.0.1"
10 )
11
12 class Computer(BaseModel):
13     computerid: int
14     cpu: Optional[str]
15     gpu: Optional[str]
16     price: float
17
18 @api.put("/computer", name = "Create a new computer")
19 def get_computer(computer: Computer):
20     """Creates a new computer within the database"""
21     return computer
```

Open URL <http://127.0.0.1:8000/docs>

My API 1.0.1 OAS 3.1

[/openapi.json](#)

My own API powered by FastAPI.

default

PUT /computer Create A New Computer

Schemas

Computer > Expand all object

HTTPValidationError > Expand all object

ValidationError > Expand all object

Edit [main.py](#) file again:

```
main.py M X
datascientest-data-engineering > fast-api > main.py > ...
1  from pydantic import BaseModel # type: ignore
2  from typing import Optional
3  from fastapi import FastAPI # type: ignore
4
5
6  api = FastAPI(
7      title = "My API",
8      description = "My own API powered by FastAPI.",
9      version = "1.0.1"
10 )
11
12 class Computer(BaseModel):
13     """a computer that is available in the store"""
14     computerid: int
15     cpu: Optional[str]
16     gpu: Optional[str]
17     price: float
18
19 @api.put("/computer", name = "Create a new computer")
20 def get_computer(computer: Computer):
21     """Creates a new computer within the database"""
22     return computer
```

Open URL <http://127.0.0.1:8000/docs> again:

My API 1.0.1 OAS 3.1

/openapi.json

My own API powered by FastAPI.

default

PUT /computer Create A New Computer

Schemas

Computer ^ Collapse all object

a computer that is available in the store

computerid* integer

cpu* ^ Collapse all (string | null)

Any of ^ Collapse all (string | null)

#0 string

#1 null

gpu* ^ Collapse all (string | null)

Any of ^ Collapse all (string | null)

#0 string

#1 null

price* number

HTTPValidationError > Expand all object

ValidationError > Expand all object

Edit [main.py](#) file again:

```
main.py M X
datascientest-data-engineering > fast-api > main.py > ...
4
5
6 api = FastAPI(
7     title = "My API",
8     description = "My own API powered by FastAPI.",
9     version = "1.0.1"
10 )
11
12 class Computer(BaseModel):
13     """a computer that is available in the store"""
14     computerid: int
15     cpu: Optional[str]
16     gpu: Optional[str]
17     price: float
18
19 @api.put("/computer", name = "Create a new computer")
20 def get_computer(computer: Computer):
21     """Creates a new computer within the database"""
22     return computer
23
24 @api.get("/custom", name = "Get custom header")
25 def get_content(custom_header: Optional[str] = Header(None, description='My own personal header')):
26     return {
27         "Custom-Header": custom_header
28     }
29
```

Open URL <http://127.0.0.1:8000/redoc>

Search...

PUT Create A New Computer

GET Get Custom Header

My API (1.0.1)

Download OpenAPI specification: [Download](#)

My own API powered by FastAPI.

Create A New Computer

Creates a new computer within the database

REQUEST BODY SCHEMA: application/json

computerid	integer (Computerid)
cpu > required	Cpu (string) or Cpu (null) (Cpu)
gpu > required	Gpu (string) or Gpu (null) (Gpu)
price > required	number (Price)

Responses

- > 200 Successful Response
- > 422 Validation Error

PUT /computer

Request samples

Payload

Content type
application/json

Copy

```
{
  "computerid": 0,
  "cpu": "string",
  "gpu": "string",
  "price": 0
}
```

Response samples

200 **422**

Content type
application/json

Copy

```
null
```

Run the following in new terminal:

```
ubuntu@ip-172-31-18-113:~$ cd datascientest-data-engineering/fast-api/
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ curl -X PUT -i
http://127.0.0.1:8000/computer \
> -H 'Content-Type: application/json' \
> -d '{
>   "computerid": 42,
>   "cpu": "Intel i9",
>   "gpu": "NVIDIA RTX 3080",
>   "price": 2499.95
> }'
HTTP/1.1 200 OK
date: Wed, 18 Jun 2025 12:44:57 GMT
server: uvicorn
content-length: 74
content-type: application/json

{"computerid":42,"cpu":"Intel i9","gpu":"NVIDIA RTX 3080","price":2499.95}ubuntu@i
p-172-31-18-113:~/datascientest-data-engineering/fast-api$
```